

Imágenes del Proyecto

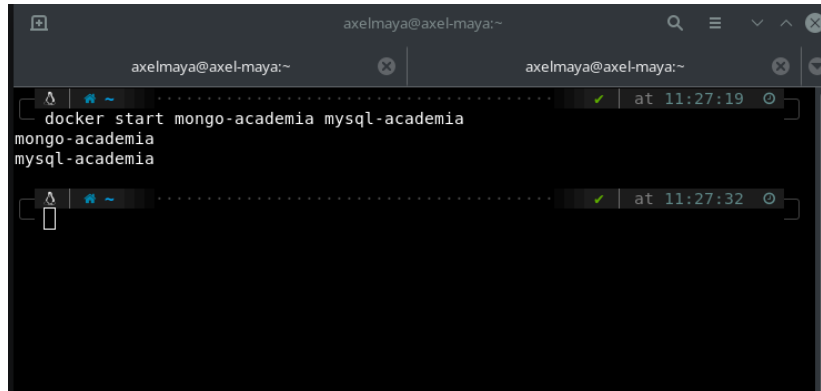


Figura 1: Contenedores Corriendo

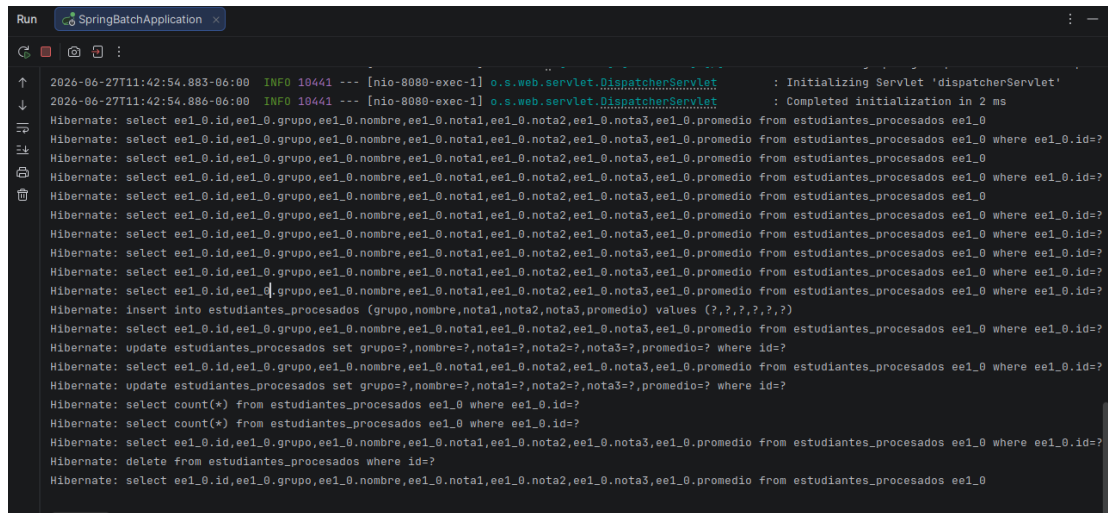


Figura 2: Compilación clase Main

```
mongosh mongodb://<credentials>@127.0.0.1:27017/?directConnection=true...
mongosh mongodb://<credentials>@127.0.0.1:... docker exec -it mysql-academia mysql -u root -p

academia> show collections
reportes
reportes_estudiantes
academia> db.reportes_estudiantes.find().pretty()
[
  {
    _id: ObjectId('6a3f4fa87e497c721a692cf1'),
    nombre: 'Juan Perez',
    grupo: 'A',
    promedio: 81.67,
    estado: 'APROBADO',
    _class: 'com.academia.batch.model.EstudianteReporte'
  },
  {
    _id: ObjectId('6a3f4fa87e497c721a692cf2'),
    nombre: 'Maria Lopez',
    grupo: 'A',
    promedio: 51.67,
    estado: 'REPROBADO',
    _class: 'com.academia.batch.model.EstudianteReporte'
  },
  {
    _id: ObjectId('6a3f4fa87e497c721a692cf3'),
    nombre: 'Carlos Garcia',
    grupo: 'B',
    promedio: 91.67,
    estado: 'APROBADO',
    _class: 'com.academia.batch.model.EstudianteReporte'
  },
  {
    _id: ObjectId('6a3f4fa87e497c721a692cf4'),
    nombre: 'Ana Martinez',
    grupo: 'B',
    promedio: 70.00
  }
]
```

Figura 3: Registros en MongoDB

```
docker exec -it mysql-academia mysql -u root -p
mongosh mongodb://<credentials>@127.0.0.1:... docker exec -it mysql-academia mysql -u root -p

mysql> show tables;
+-----+
| Tables_in_academia |
+-----+
| BATCH_JOB_EXECUTION |
| BATCH_JOB_EXECUTION_CONTEXT |
| BATCH_JOB_EXECUTION_PARAMS |
| BATCH_JOB_EXECUTION_SEQ |
| BATCH_JOB_INSTANCE |
| BATCH_JOB_SEQ |
| BATCH_STEP_EXECUTION |
| BATCH_STEP_EXECUTION_CONTEXT |
| BATCH_STEP_EXECUTION_SEQ |
| empleados_procesados |
| estudiantes_procesados |
+-----+
11 rows in set (0.01 sec)

mysql> select * from estudiantes_procesados;
+-----+
| id | nombre | grupo | nota1 | nota2 | nota3 | promedio |
+-----+
| 12 | Maria Lopez | A | 60.00 | 55.00 | 40.00 | 51.67 |
| 13 | Carlos Garcia | B | 95.00 | 88.00 | 92.00 | 91.67 |
| 14 | Ana Martinez | B | 70.00 | 72.00 | 68.00 | 70.00 |
| 15 | Pedro Sanchez | A | 50.00 | 45.00 | 60.00 | 51.67 |
| 16 | Laura Diaz | C | 85.00 | 90.00 | 80.00 | 85.00 |
| 17 | Roberto Flores | C | 40.00 | 55.00 | 50.00 | 48.33 |
| 18 | Sofia Ramirez | B | 100.00 | 95.00 | 98.00 | 97.67 |
| 19 | Diego Torres | A | 65.00 | 70.00 | 75.00 | 70.00 |
| 20 | Fernanda Rios | C | 72.00 | 68.00 | 74.00 | 71.33 |
| 21 | Nuevo Alumno | A | 80.00 | 85.00 | 90.00 | 85.00 |
+-----+
```

Figura 4: Registros en MySQL

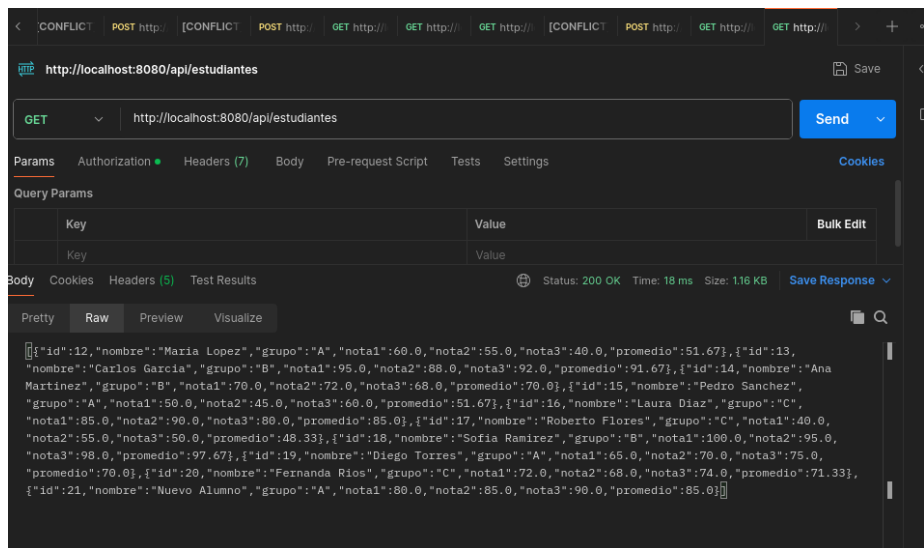


Figura 5: Petición GET en Postman

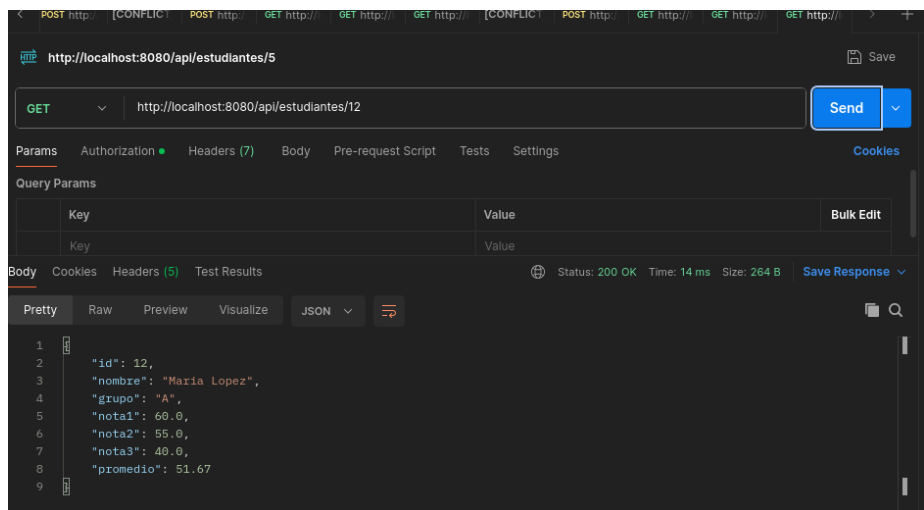


Figura 6: Petición GET/1 en Postman

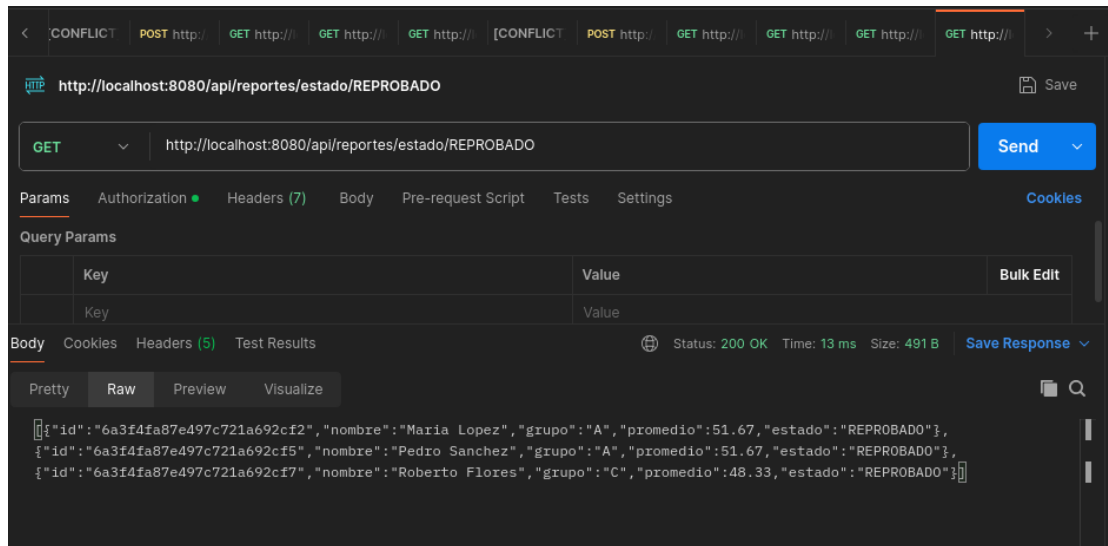


Figura 7: Obteniendo Reprobados

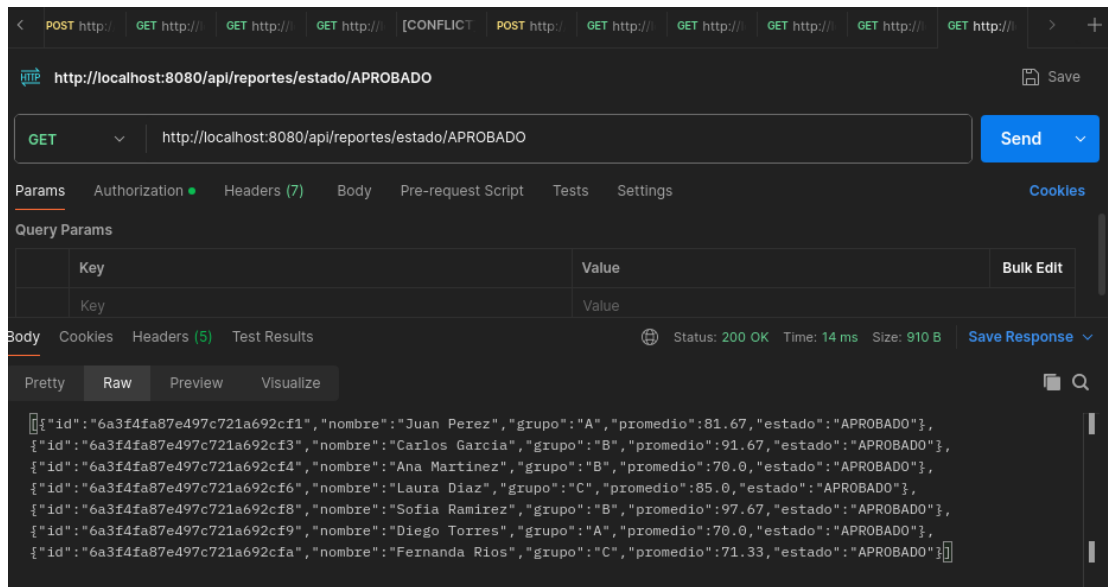
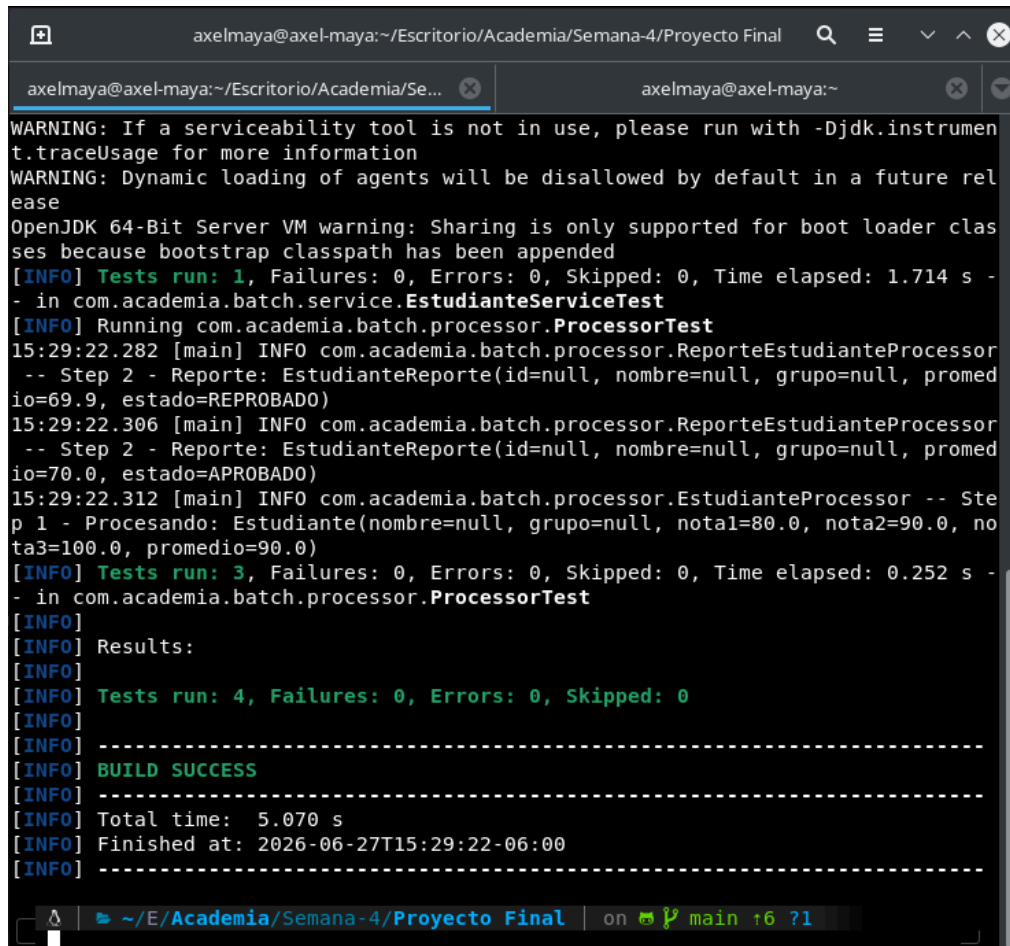


Figura 8: Obteniendo Aprobados

```
axelmaya@axel-maya:~  
mongosh mongodb://<credentials>@127.0.0.1:27017  
axelmaya@axel-maya:~  
took 11m 12s at 03:22:00  
curl -i -X DELETE http://localhost:8080/api/estudiantes/12  
HTTP/1.1 204  
Date: Sat, 27 Jun 2026 21:22:18 GMT  
axelmaya@axel-maya:~  
at 03:22:18  
curl http://localhost:8080/api/estudiantes/aprobados/total  
{"aprobados":7}  
axelmaya@axel-maya:~  
at 03:22:32  
curl http://localhost:8080/api/reportes  
{  
  "id": "6a3f4fa87e497c721a692cf1", "nombre": "Juan Perez", "grupo": "A", "promedio": 85.67, "estado": "APROBADO",  
  "id": "6a3f4fa87e497c721a692cf2", "nombre": "Maria Lopez", "grupo": "A", "promedio": 51.67, "estado": "REPROBADO",  
  "id": "6a3f4fa87e497c721a692cf3", "nombre": "Carlos Garcia", "grupo": "B", "promedio": 91.67, "estado": "APROBADO",  
  "id": "6a3f4fa87e497c721a692cf4", "nombre": "Ana Martinez", "grupo": "B", "promedio": 70.0, "estado": "APROBADO",  
  "id": "6a3f4fa87e497c721a692cf5", "nombre": "Pedro Sanchez", "grupo": "A", "promedio": 51.67, "estado": "REPROBADO",  
  "id": "6a3f4fa87e497c721a692cf6", "nombre": "Laura Diaz", "grupo": "C", "promedio": 85.0, "estado": "APROBADO",  
  "id": "6a3f4fa87e497c721a692cf7", "nombre": "Roberto Flores", "grupo": "C", "promedio": 48.33, "estado": "REPROBADO",  
  "id": "6a3f4fa87e497c721a692cf8", "nombre": "Sofia Ramirez", "grupo": "B", "promedio": 97.67, "estado": "APROBADO",  
  "id": "6a3f4fa87e497c721a692cf9", "nombre": "Diego Torres", "grupo": "A", "promedio": 70.0, "estado": "APROBADO",  
  "id": "6a3f4fa87e497c721a692cfa", "nombre": "Fernanda Rios", "grupo": "C", "promedio": 71.33, "estado": "APROBADO"}  
axelmaya@axel-maya:~  
at 03:22:48  
curl -i -X PATCH http://localhost:8080/api/estudiantes/11 \\  
-H "Content-Type: application/json" \\  
-d '{"grupo": "Z"}'  
HTTP/1.1 404  
Content-Length: 0  
Date: Sat, 27 Jun 2026 21:23:05 GMT
```

Figura 9: Otras Peticiones



```
axelmaya@axel-maya:~/Escritorio/Academia/Semana-4/Proyecto Final
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
WARNING: Dynamic loading of agents will be disallowed by default in a future release
OpenJDK 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been appended
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.714 s - in com.academia.batch.service.EstudianteServiceTest
[INFO] Running com.academia.batch.processor.ProcessorTest
15:29:22.282 [main] INFO com.academia.batch.processor.ReporteEstudianteProcessor -- Step 2 - Reporte: EstudianteReporte(id=null, nombre=null, grupo=null, promedio=69.9, estado=REPROBADO)
15:29:22.306 [main] INFO com.academia.batch.processor.ReporteEstudianteProcessor -- Step 2 - Reporte: EstudianteReporte(id=null, nombre=null, grupo=null, promedio=70.0, estado=APROBADO)
15:29:22.312 [main] INFO com.academia.batch.processor.EstudianteProcessor -- Step 1 - Procesando: Estudiante(nombre=null, grupo=null, nota1=80.0, nota2=90.0, nota3=100.0, promedio=90.0)
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.252 s - in com.academia.batch.processor.ProcessorTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 5.070 s
[INFO] Finished at: 2026-06-27T15:29:22-06:00
[INFO] -----
```

Figura 10: Resultado de tests